

2.4 – Transistors



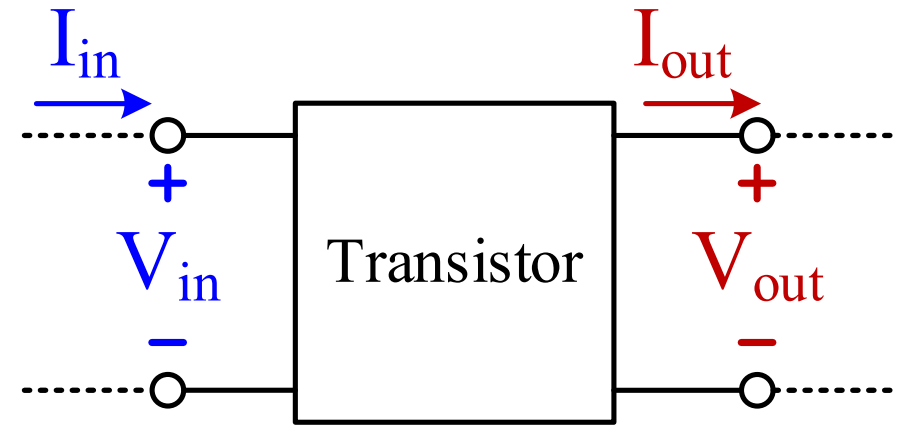
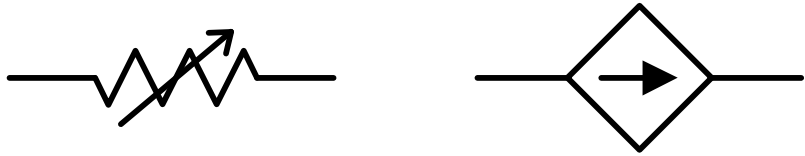
2.4.1 – Transistor Operation and Circuit Models

Transistors have Controllable Flow

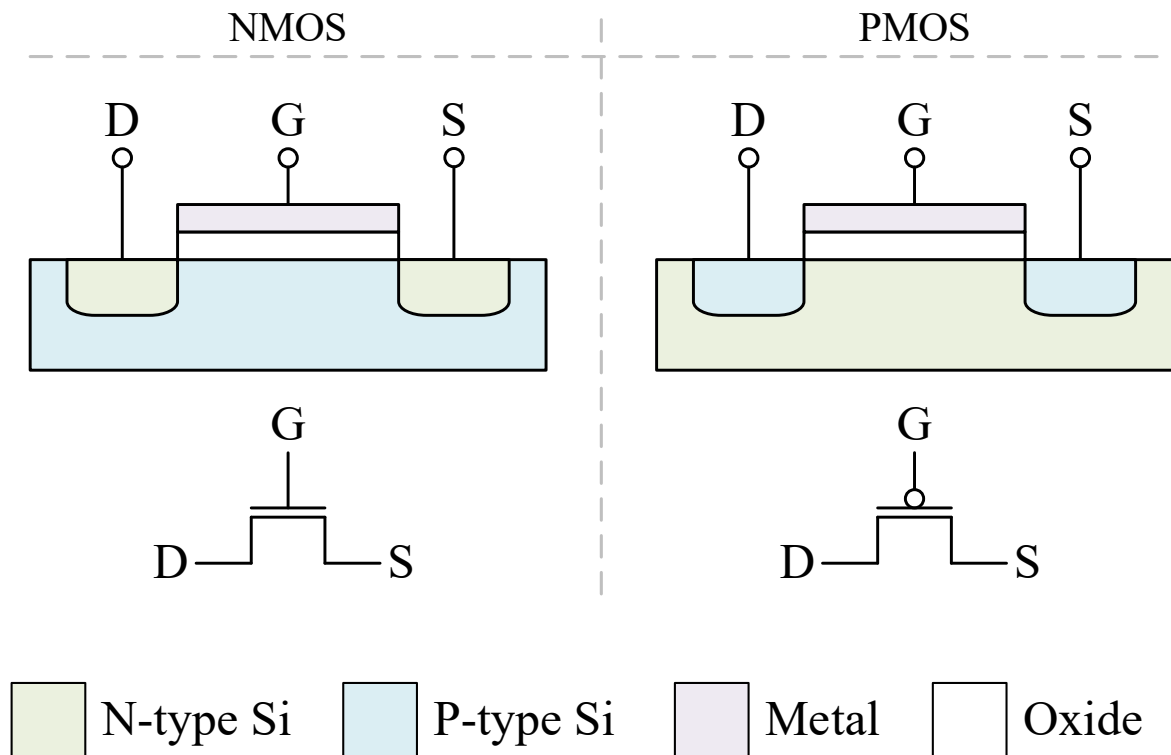
- I_{in} (BJT) or V_{in} (MOSFET) determines I_{out} - V_{out} relationship
- A small change in input can cause a big change in output
- Use as controllable switch (digital)



- Use as controllable resistor or current source (analog)



MOSFET Details and Structure

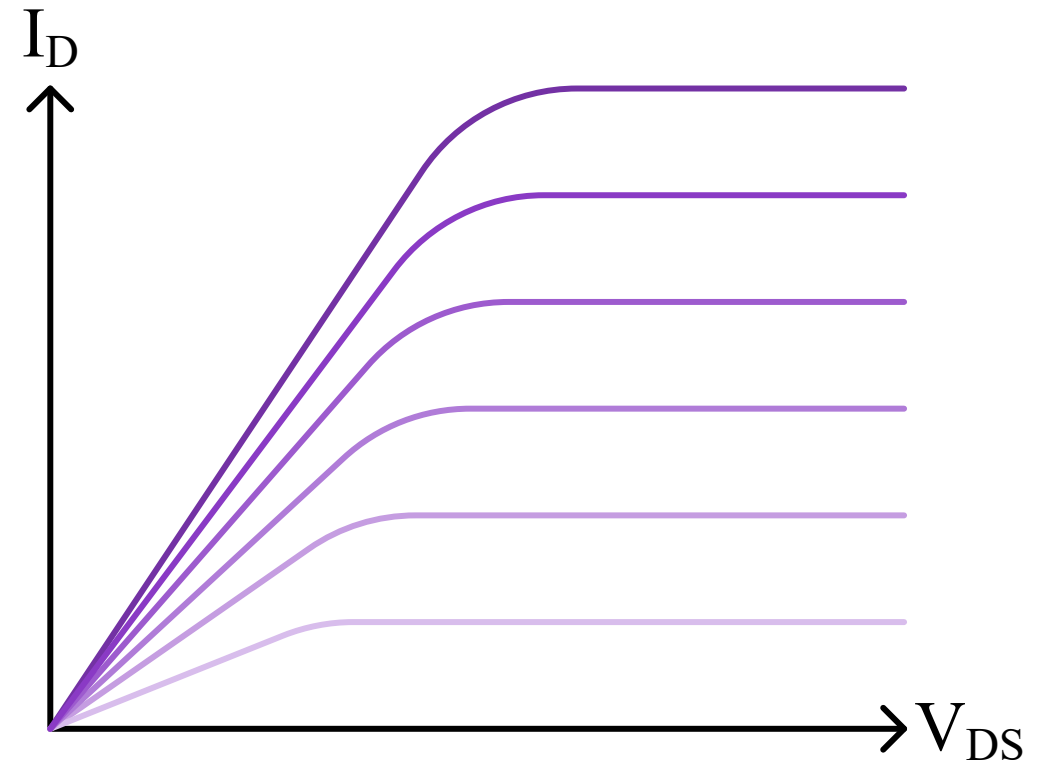
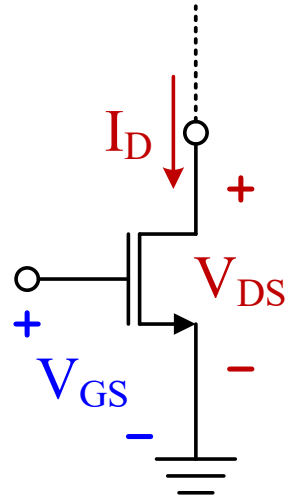
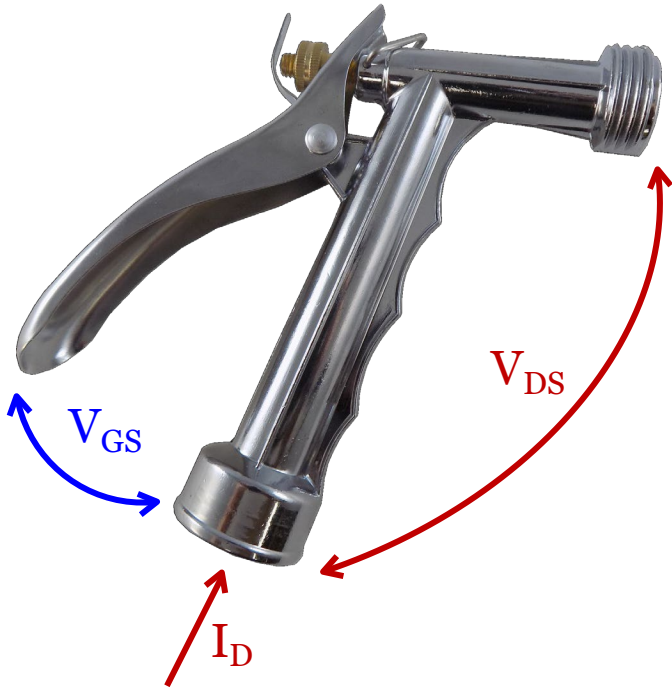


- MOSFET = Metal Oxide Semiconductor Field Effect Transistor
- Three-terminals: Gate, Source, Drain
 - Gate voltage controls Drain to Source conduction
- Two flavours
 - P-type uses holes (h^+) to conduct
 - N-type uses electrons (e^-) to conduct

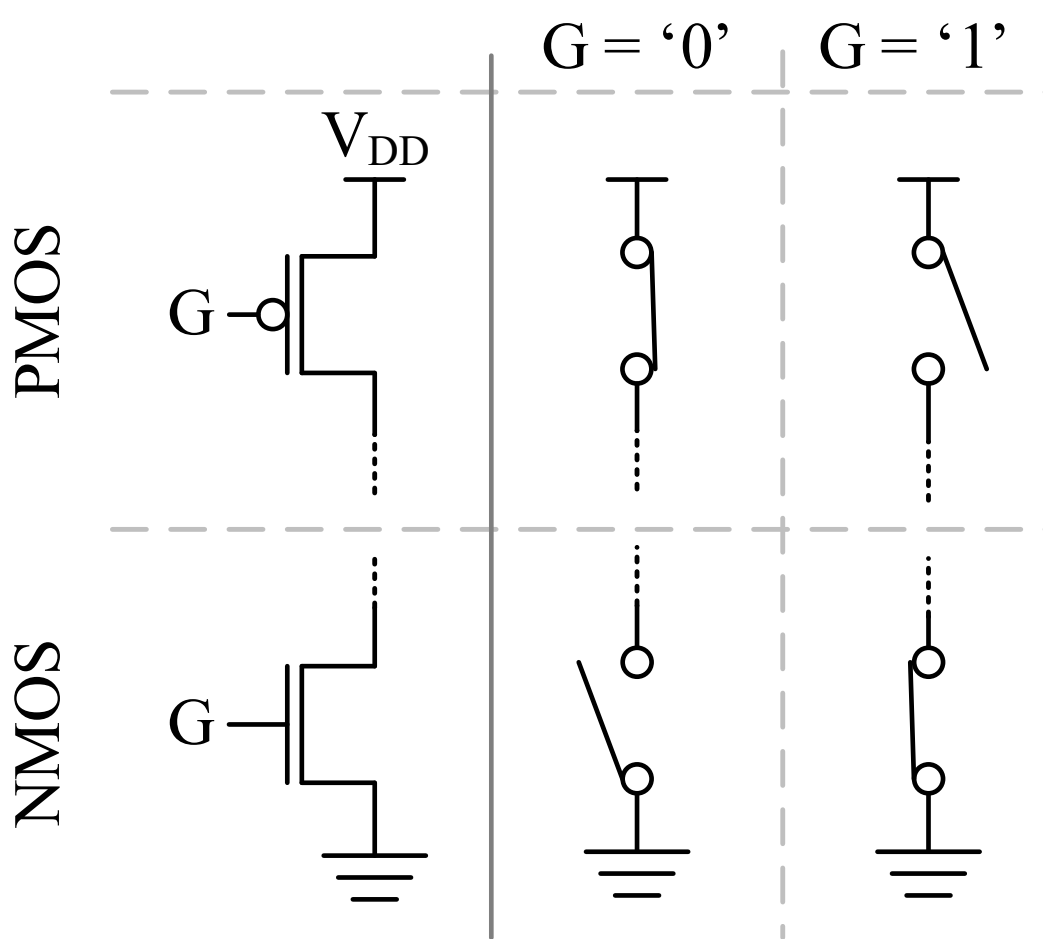


MOSFET Nozzle Analogy

- V_{GS} controls flow – like squeezing a nozzle
- V_{DS} is the hose pressure
- I_D is the resulting flow rate



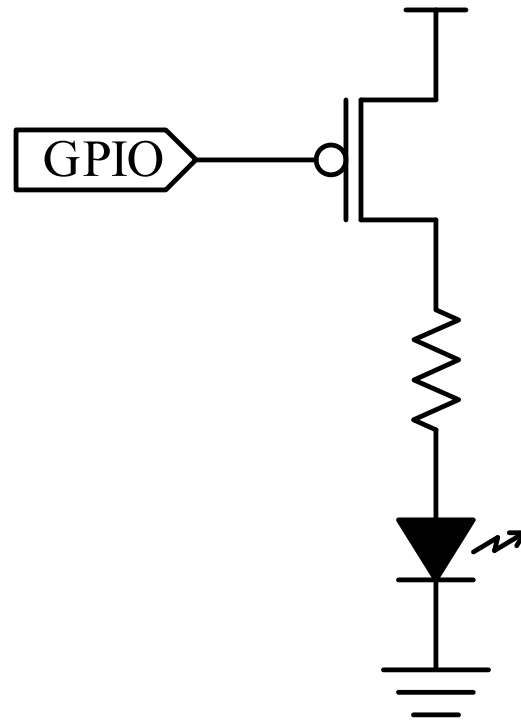
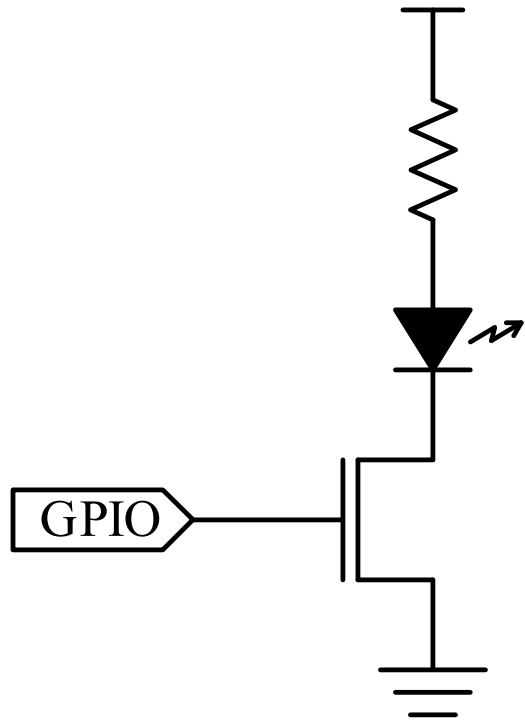
MOSFET Switch Model (Digital)



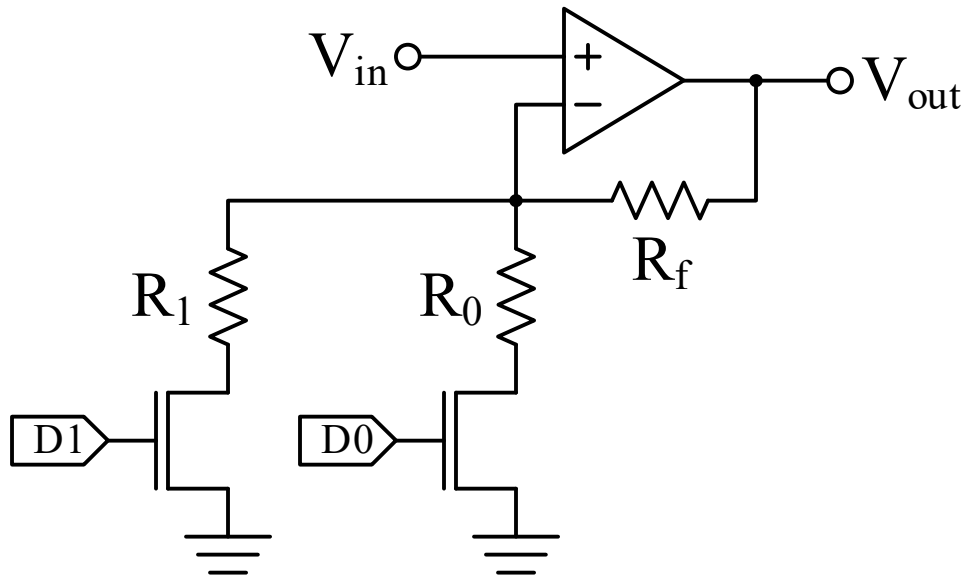
- NMOS is good at “pulling down” to GND
- PMOS is good at “pulling up” to V_{DD}
- They will accidentally shut themselves off if used opposite
- May include ON resistance: $R_{DS,ON}$



Example: Turning on an LED



Example: Programmable Gain Amplifier



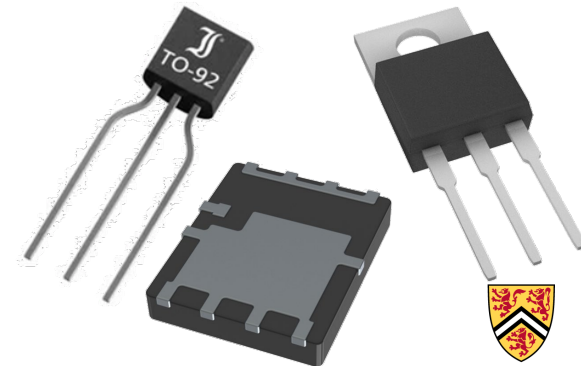
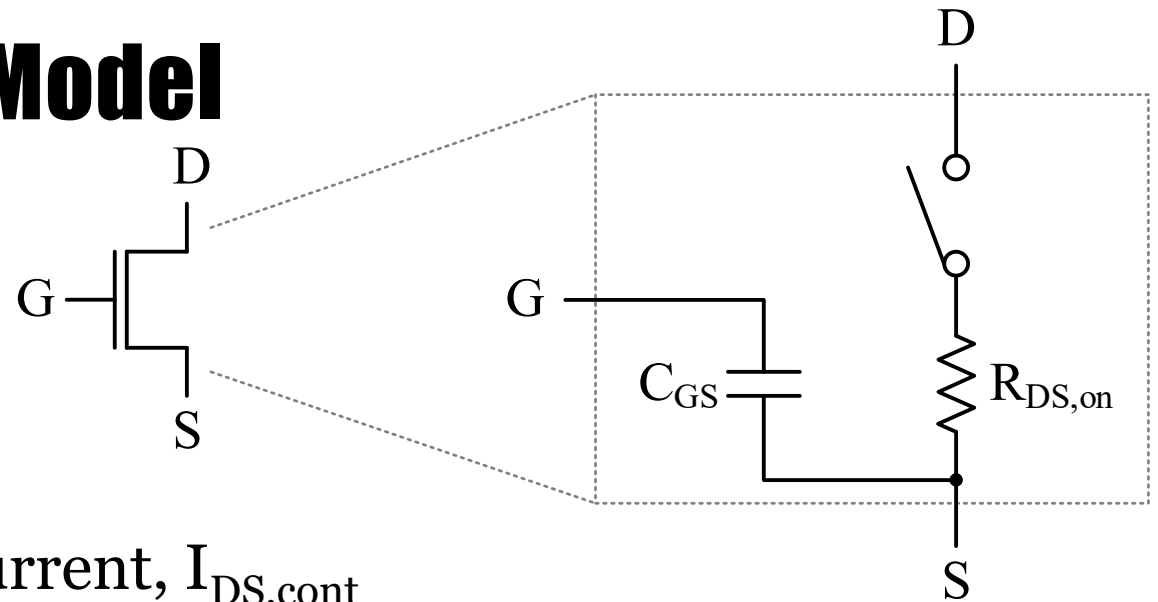
Determine the possible gains when $R_f = R_o = 10 \text{ k}\Omega$ and $R_1 = 5 \text{ k}\Omega$

D1	D0	R_i	Gain [V/V]
0	0	o.c.	
0	1	R_o	
1	0	R_1	
1	1	$R_1 R_o$	



Simple Digital MOSFET Circuit Model

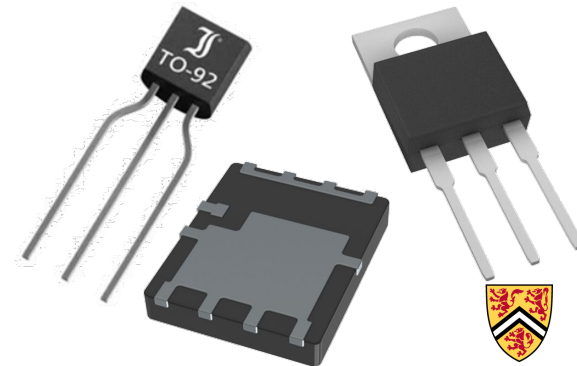
- Gate input is a capacitor, C_{GS}
- Minimum drain-to-source “ON” resistance, $R_{DS,ON}$
- Maximum continuous drain-to-source current, $I_{DS,cont}$
- Examples:
 - 2N7000 (\$0.40): $C_{GS} = 20 \text{ pF}$, $R_{DS,on} = 1.2 \Omega$, $I_{DS,cont} = 110 \text{ mA}$
 - IRF610 (\$1.95): $C_{GS} = 140 \text{ pF}$, $R_{DS,on} = 1.5 \Omega$, $I_{DS,cont} = 3.3 \text{ A}$
 - IQDH29 (\$4.55): $C_{GS} = 13 \text{ nF}$, $R_{DS,on} = 0.2 \text{ m}\Omega$, $I_{DS,cont} = 789 \text{ A}$



MOSFET Design Choices

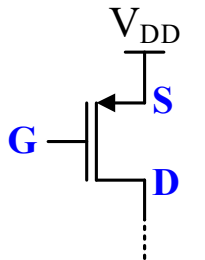
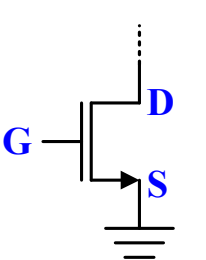
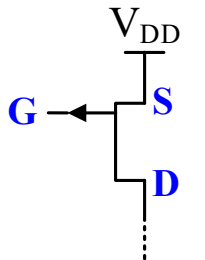
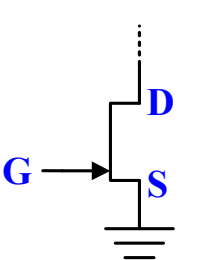
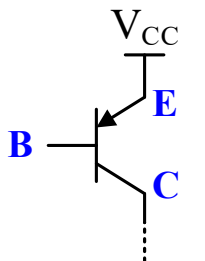
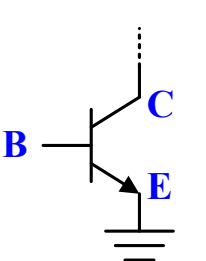
Which would you choose for the following scenarios?

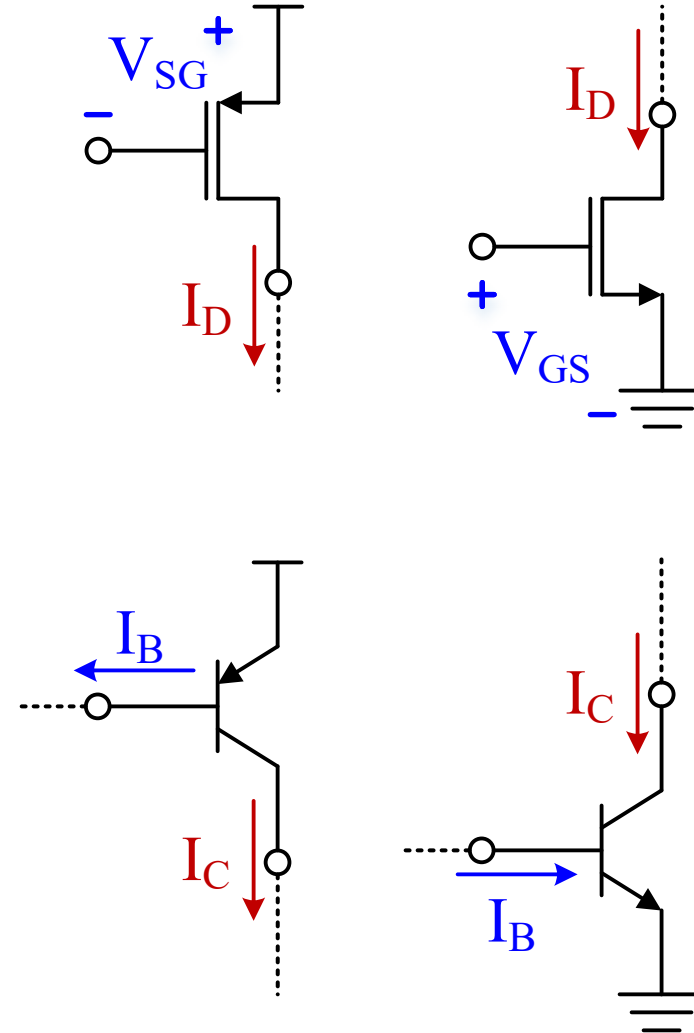
1. Your system occasionally powers a cooling fan that uses a 5 V DC motor (2 A max current)
 2. You are developing a high-current eBike fast charger
 3. You need to toggle a transistor ON and OFF as fast as possible
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- A. 2N7000 (\$0.40): $C_{GS} = 20 \text{ pF}$, $R_{DS,on} = 1.2 \Omega$, $I_{DS,cont} = 110 \text{ mA}$
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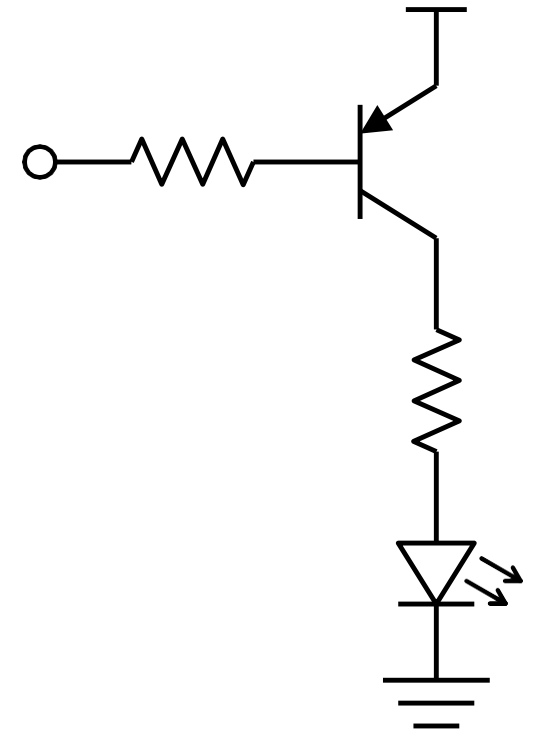
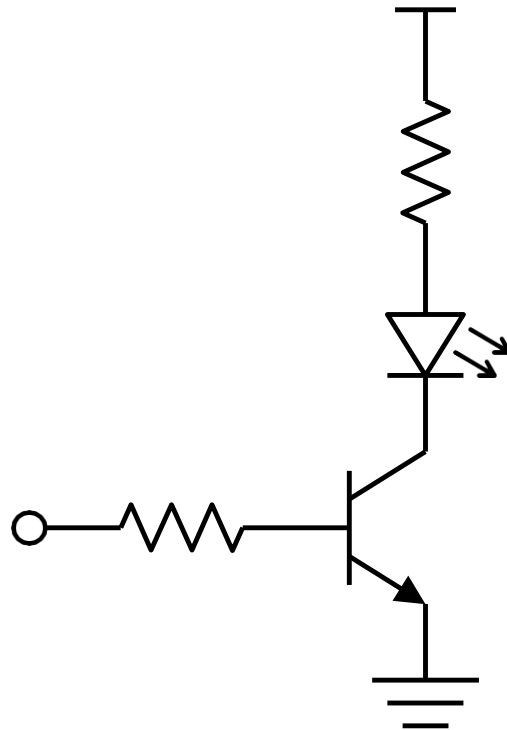
2.4.2 – Transistors Types

Types of Transistors

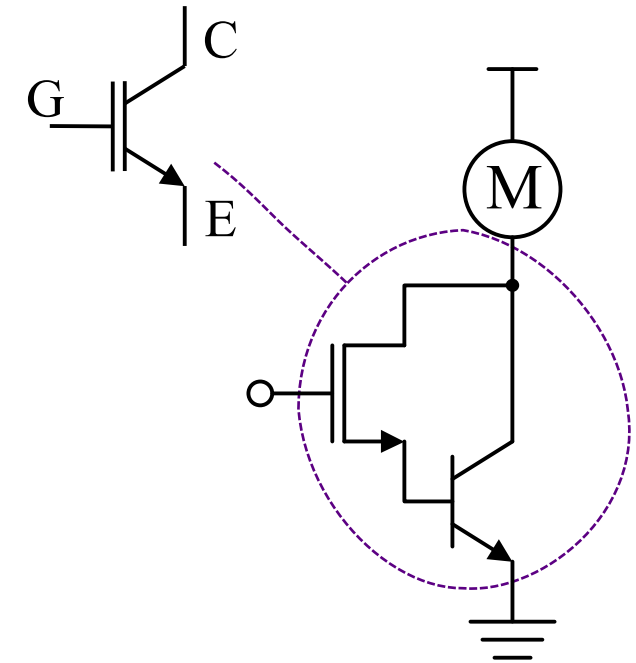
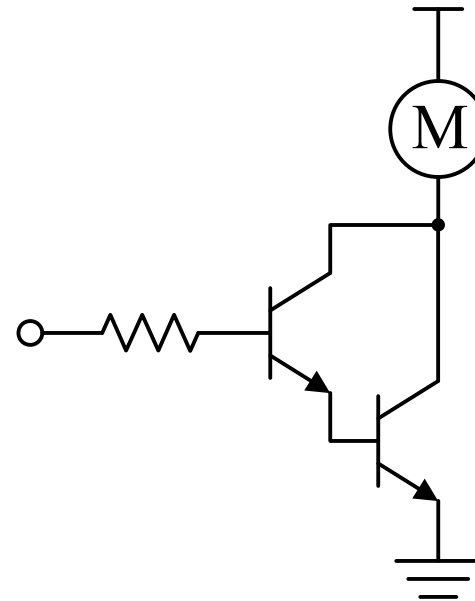
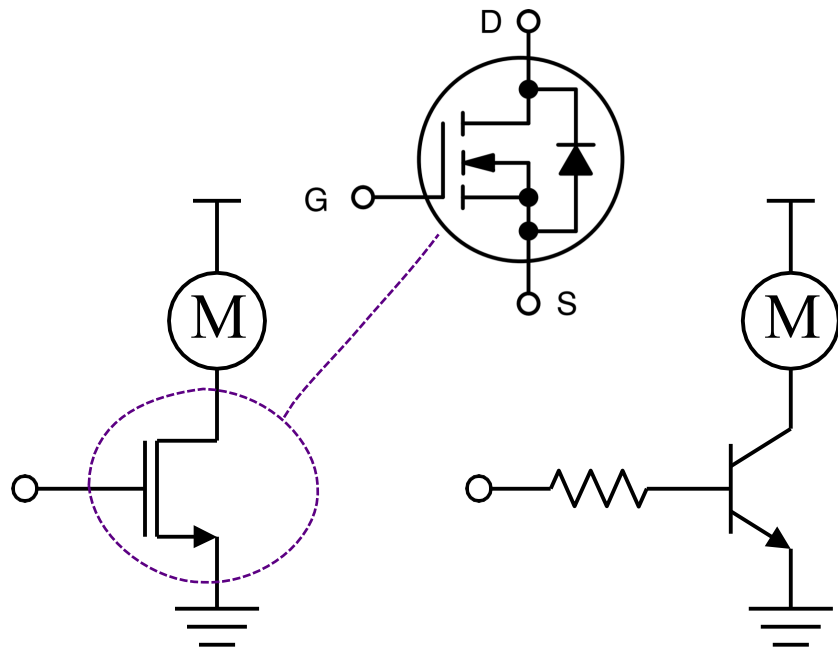
	P-Type	N-Type	Notes
MOSFET			
JFET			
BJT			



Using a BJT



Example: Driving a Large Current

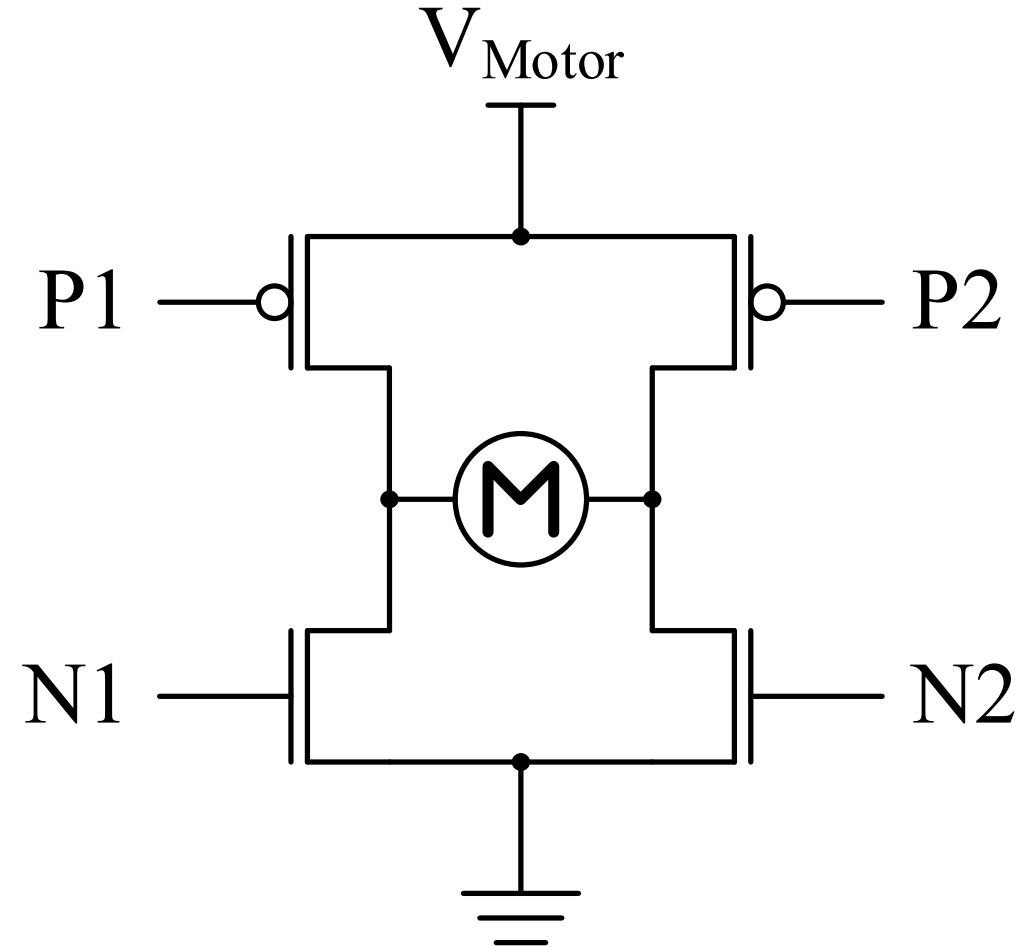


2.4.3 – Transistor Circuits

Digital: H-Bridge Circuits

- Steers current in either direction
- Complete with '1' = V_{Motor} , '0' = GND

Mode	P1	P2	N1	N2
Forward				
Reverse				
Coast				
Brake				
Short Circuit				
Short Circuit				

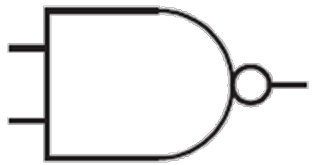


- Can be used whenever bidirectional current is required

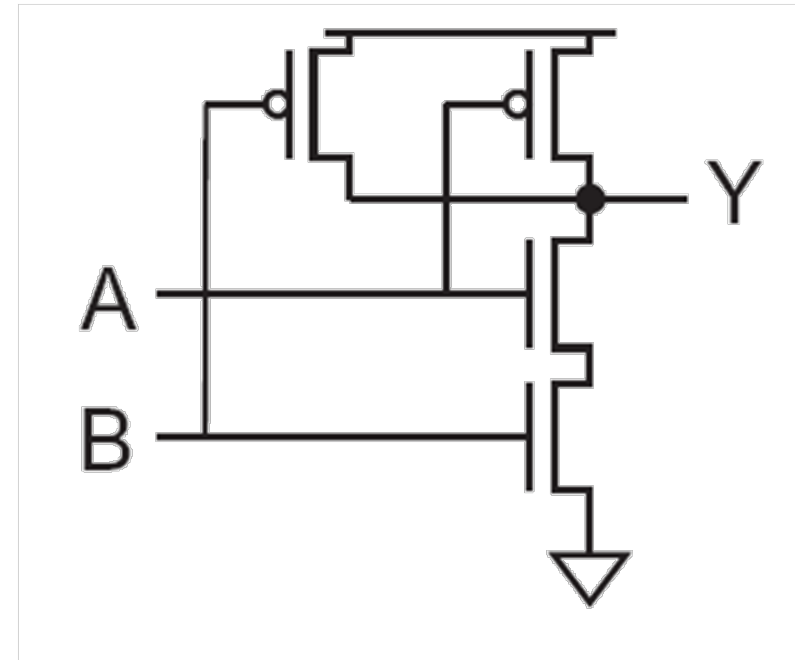


Digital CMOS

- PMOS pulls-up to output '1'
- NMOS pulls down to output '0'
- E.g., 2-input NAND gate

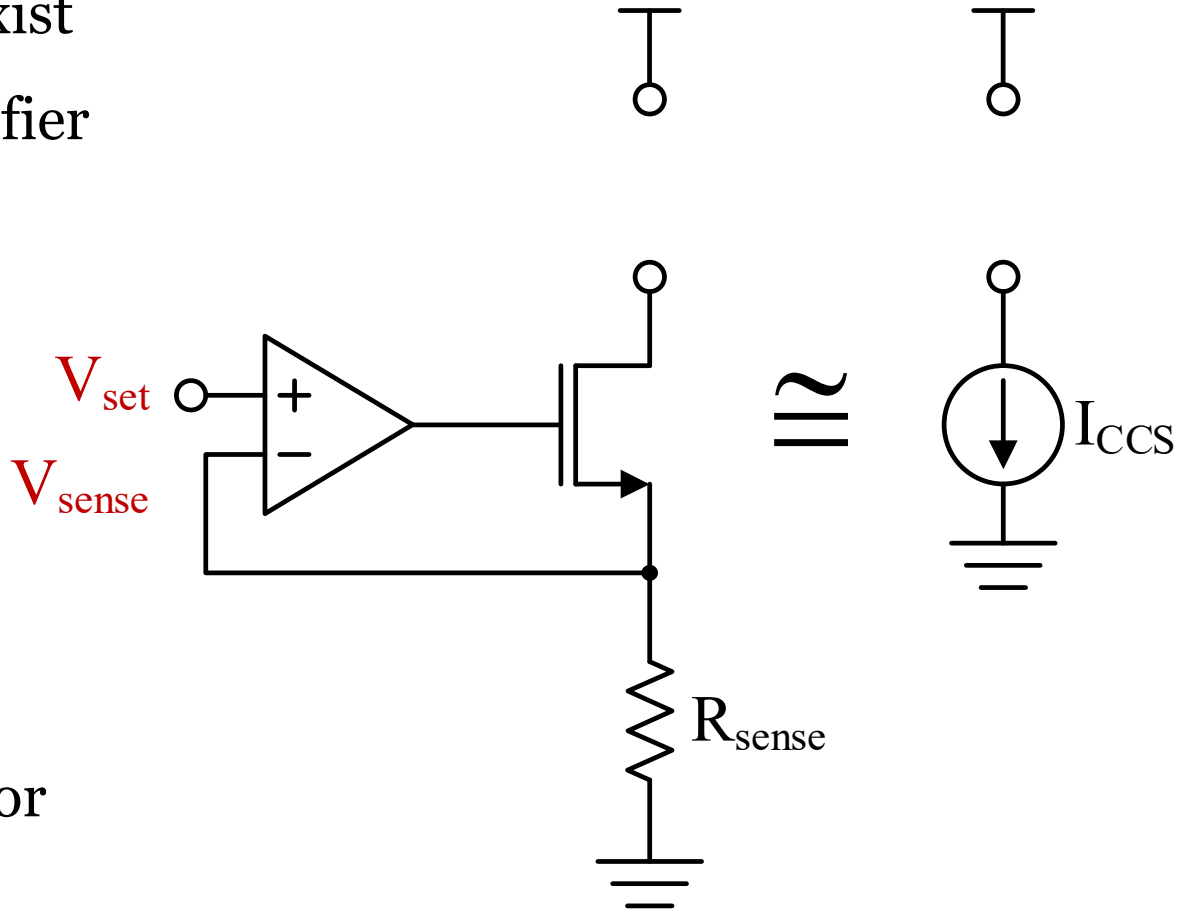


A	B	Pull-Down Network	Pull-Up Network	Y
0	0	OFF	ON	1
0	1	OFF	ON	1
1	0	OFF	ON	1
1	1	ON	OFF	0



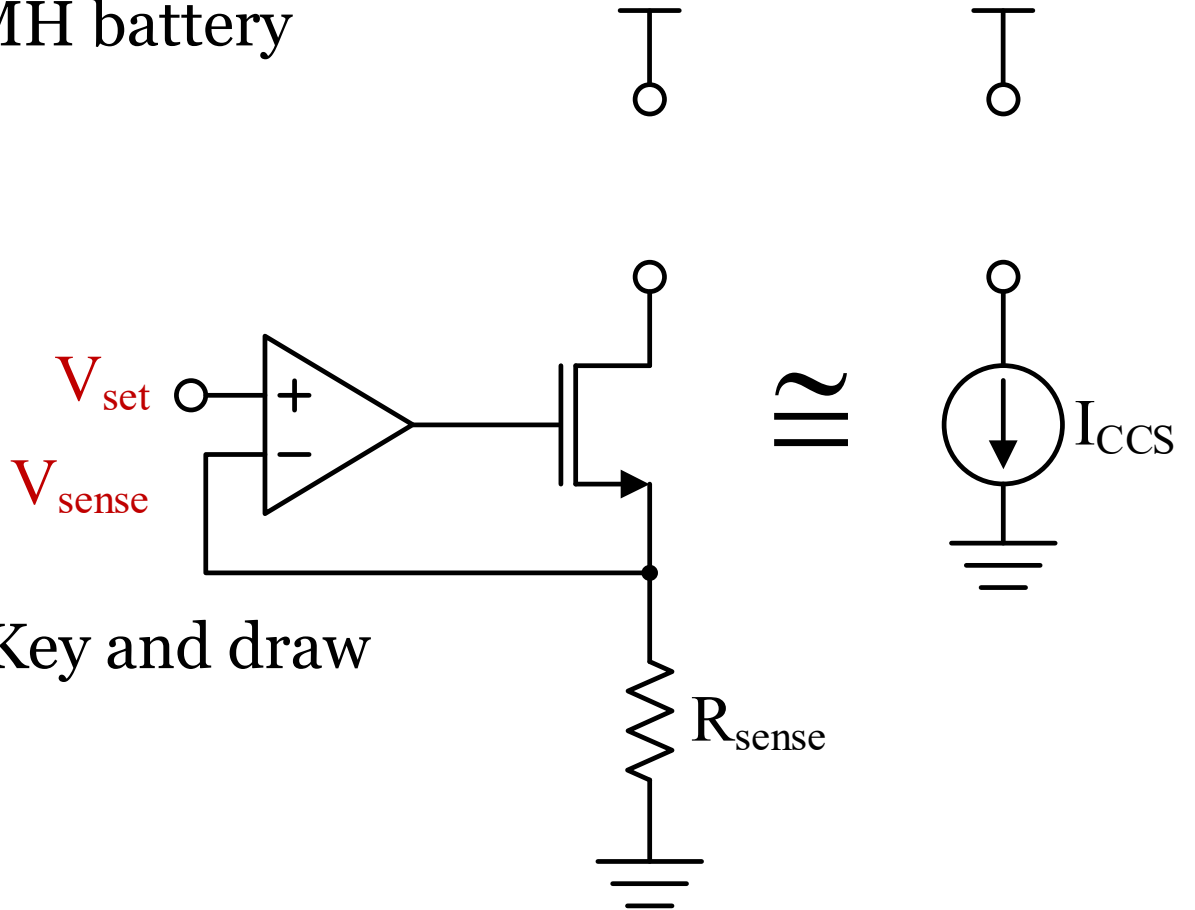
MOSFETs as Variable Resistors: Controlled Current Source

- Constant Current source does not actually exist
- Make with a current-sensing feedback amplifier
 - Sense current using voltage across a low-value resistor (to save on I^2R losses)
 - Control MOSFET gate voltage to act as variable resistor
 - Negative feedback to maintain desired current
- Called OTA: Operational Transconductance Amplifier [A/V]
- Used in: Battery charging, LED drivers, motor torque control, electrochemical reactions



Controlled Current Source Example

- Design this circuit to charge a 14.4 V NiMH battery
- Given:
 - I_{CCS} from 0 to 2 A
 - An MCU DAC can generate V_{set} from 0 to 3.3 V
 - $V_{DD} = 18\text{ V}$
- Choose R_{sense} and a MOSFET from Digi-Key and draw the completed circuit.

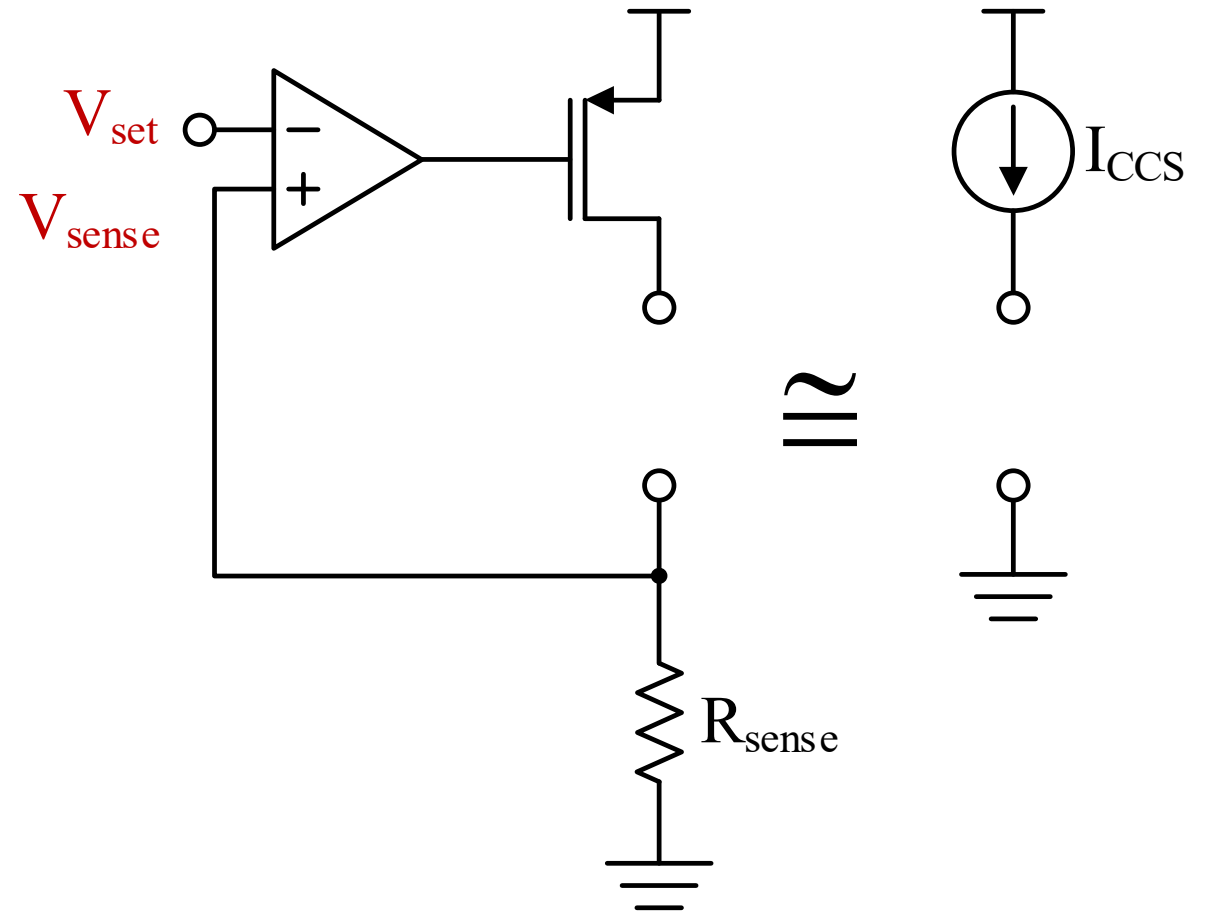


Controlled Current Source Example



PMOS Controlled Current Source

- Previous NMOS circuit sinks current to GND (Current Sink)
 - Recall NMOS good at “pulling down” to GND but not “pulling up” to VDD
 - Also note exposed terminal connected to VDD (possible safety issue)
- Current Source requires PMOS
 - PMOS inverted action flips feedback path (still negative feedback)
 - Exposed terminal is ~GND
 - PMOS less efficient than NMOS, but still usually okay



LTSpice Modeling Challenge



- Implement a battery charging circuit
- Start with the NMOS Controlled Current Source example
 - Modify it for a battery of your choosing (find the charging requirements!)
- Model the battery as a large capacitor (voltage will drop with discharge):
 - $C_{eq} = 3600B / (V_{max} - V_{min})$
 - C_{eq} is the equivalent capacitance in F
 B is the battery capacity in A·h
 $V_{max} - V_{min}$ is the difference between charged and discharged battery voltages in V
 - E.g., a 14.4 V NiMH battery might have $B = 3$ Ah, $V_{max} - V_{min} = 17.4 - 12.0 = 5.4$ V, giving $C_{eq} = 2000$ F
 - You will need to set the capacitor's initial voltage for the simulation
- Try improving it: Can you get the battery negative terminal grounded? Can you detect and protect against a short circuit? Can you implement a realistic charging cycle recommended by the battery manufacturer? What other improvements can you make?

